

Course Outline

Cybersecurity – LEA.D8

A. General Information

Program Name	Cybersecurity (CYB)
Course title	Cyber Defense
Course and section number	420-D03-AB
Ponderation <i>Ratio of lecture, practical and homework hours</i>	3-3-3
Number of Credits	3.00
Competency statement(s) and code(s)	EZ03 To apply access control measures. Covered: Completely ☒ Partially ☐
Prerequisite (s)	420-D01-AB Preventive Surveillance in Cybersecurity
Modality (or course delivery)	Online
Semester	TBA
Teacher name(s)	TBA
Teacher's contact info	Email address
Cohort	CYB-01
Start date	
End date	
Day(s) and times	
Classroom/lab number	

B. Introduction

This course is the third course of the Cybersecurity intensive program leading to an *Attestation d'études collégiales* (A.E.C.).

This course enables students to secure and defend an IT infrastructure against malicious actors internal or external to the organization. From network architecture to privilege and access management, active defense techniques and cryptography to protect information and ensure business continuity in case of a cybersecurity incident. The students will learn how to assess and increase the resilience of the network and its components to different types of cyberattacks.

The skills acquired in this course will be reused in the Intrusion Testing and Cybersecurity Incident Management courses, where access control measures are directly tested.

The topics covered in this course are

- Install and configure network defense measures on Windows servers
- Install and configure network defense measures on Linux servers
- Configure a network firewall for both wired and wireless networks
- Assess and document network resilience.

Students will be instructed with regards to the use of on-premises equipment as well as cloud platforms such as Amazon Web Services and Microsoft Azure to defend an IT infrastructure for a typical company.

C. Course Objectives

Upon successful completion of this course, students will be able to:

EZ03	
Statement of the Competency	Achievement Context
Apply access control measures.	▪ For an entire network with various platforms ▪ In response to an ad hoc request or a service requisition ▪ Autonomously ▪ In collaboration with analysts and other technicians ▪ In accordance with internal practices, procedures and policies ▪ Based on a risk analysis ▪ Using: – Technical manuals – Documents describing the infrastructure – Hardware, tools and software.
Performance Criteria for the Competency as a Whole	
	• Compliance with the rules and regulations • Compliance with professional ethics • Demonstration of autonomy • Effective management of priorities • Ability to solve problems • Compliance with policies and standards • Respect for operational constraints • Compliance with good cybersecurity practices • Consideration of relevant sources of information • Effective work methods • Appropriate use of relevant hardware, tools and software • Detailed documentation of the operations carried out
Elements of the Competency	Performance Criteria
1. Apply cyberdefence techniques. ¹	1.1 Accurate identification of perimeter elements 1.2 Establishment of appropriate security measures on the perimeter 1.3 Relevant subdivision of the network into zones 1.4 Proper configuration of active defence systems 1.5 Application of relevant redundancy measures

¹ In the context of this AEC, the term cyber defense refers to all the techniques and actions that make an infrastructure defensible and ensure its cyber resilience. Cyber defense techniques include active defense techniques, perimeter security, defense in depth, and redundancy elements that enable business continuity in the event of a cyber attack.

2. Control access to hosts.	2.1 Proper identification of the position, role and nature of the host in the infrastructure 2.2 Effective control of digital identities, permissions and access 2.3 Compliant configuration of a system firewall 2.4 Application of relevant control measures depending on the role of the system 2.5 Methodical application of hardening measures 2.6 Rigorous validation of the effectiveness of the access control measures applied
3. Control network access.	3.1 Effective control of digital identities, permissions and access 3.2 Determination of appropriate access to the different zones of the network 3.3 Compliant configuration of a network firewall 3.4 Proper implementation of intrusion prevention and detection systems 3.5 Proper use of security protocols on wired and wireless networks 3.6 Rigorous validation of the effectiveness of the access control measures applied
4. Apply cryptography techniques.	4.1 Accurate application of the basic concepts of cryptography 4.2 Compliant implementation of the main cryptosystems 4.3 Summary analysis of the main attacks on cryptographic systems 4.4 Accurate assessment of the relative effectiveness of various cryptographic systems, depending on the application 4.5 Use of appropriate techniques to secure transient and stored data

D. Course Content and Schedule

Cyberdefense fundamentals and perimeter security

- Defense methods, techniques, and tools for securing IT infrastructures
- Defense-in-depth and cyber resilience concepts
- Identification of network entry points and common cyberattack vectors
- Defensible network architecture design and zoning for access control
- Critical security controls defined by the CIS (Center for Internet Security)
- Tools for monitoring and securing traffic between zones with different trust levels
- Role and function of a Security Operations Center (SOC)
- Assessing and documenting network resilience

Network access control and security

- Privilege and access management on Windows and Linux systems
- Processes and tools to implement “need-to-know” access control

- Secure configuration of host-based firewalls
- Techniques for securing access to sensitive data and systems
- Countering threats: backdoors, Trojans, rootkits, worms, bots, botnets
- Malware detection and prevention measures

Securing Windows and Linux hosts

- Managing Windows accounts, groups, and security policies via Active Directory
- Applying hardening measures to Windows and Linux hosts
- Monitoring, controlling, and remediating vulnerabilities through configuration management
- Validation of host security configurations

Intrusion Detection and Prevention Systems (IDS/IPS)

- Architecture and operation of network-based and host-based IDS/IPS
- Deployment of IDS/IPS at network access points and endpoints
- Web application security: injection attacks and countermeasures
- Critical security controls and their implementation in active defense

Cyber defense tools and techniques

- Network device security: securing routers, switches, and wireless networks
- Implementation of network firewalls: configuration for wired and wireless environments
- Secure communication protocols (e.g., SSH, HTTPS, SSL/TLS, IPsec)
- Network software update and patch management
- Securing network configurations through device registry and managements plans
- Security considerations for virtual platforms and cloud services (AWS, Azure)
- Web application and service security, including common vulnerabilities
- Techniques and tools for session hijacking prevention and cache protection
- Denial-of-Service (DoS) attack defense strategies

Cryptography and secure communication

- Symmetric and asymmetric cryptography algorithms, hash functions, steganography
- Implementation of cryptographic measures for securing data in transit and at rest
- Public Key Infrastructure (PKI) and digital certificates
- Secure storage and hashing of credentials (passwords)
- Detection and defense against attacks targeting cryptosystems
- Secure VPN access configuration
- Secure services configuration (SSH, HTTPS, SSL, TLS)
- Processes for securing cloud-based and local storage systems using encryption

Topics:	Day /Dates	Additional Info:	Online	In-person
Course overview Cyberdefense fundamentals and perimeter security	1 Sat. Aug. 29	Introduction to course, defense methods, zoning concepts – LAB_01	☒	
Cyberdefense fundamentals and perimeter security	2 Tue. Sept. 1	Zoning concepts and Q_01	☒	
Network access control and security	3 Wed. Sept. 2	Privilege management – LAB_02	☒	
Network access control and security	4 Tue. Sept. 8	Windows and Linux access control – LAB_02	☒	
Securing Windows and Linux hosts	5 Wed. Sept. 9	Host hardening techniques – LAB_03	☒	
Securing Windows and Linux hosts	6 Sat. Sept. 12	Host firewall configuration – LAB_03	☒	
Intrusion Detection and Prevention Systems (IDS/IPS)	7 Tue. Sept. 15	IDS and IPS deployment – LAB_04	☒	
Intrusion Detection and Prevention Systems (IDS/IPS)	8 Wed. Sept. 16	IDS and IPS configuration and validation – LAB_04	☒	
Mid-Term review	9 Sat. Sept. 19	Q_02 and Mid-Term Review	☒	
Mid-Term Exam	10 Tue. Sept. 22	MID-TERM EXAM	☒	
Cyber defense tools and techniques	11 Wed. Sept. 23	Network device security and secure communication protocols – LAB_05	☒	
Cyber defense tools and techniques	12 Sat. Sept. 26	Network firewalls – LAB_05	☒	
Cyber defense tools and techniques	13 Tue. Sept. 29	Wi-Fi and cloud security practices	☒	
Cryptography and secure communication	14 Wed. Sept. 30	Cryptography fundamentals and hashing, and Q_03	☒	

Topics:	Day /Dates	Additional Info:	Online	In-person
Cryptography and secure communication	15 Tue. Oct. 6	PKI concepts	☒	
Cryptography and secure communication	16 Wed. Oct. 7	LAB_06: VPN setup and encryption configuration (1)	☒	
Cryptography and secure communication	17 Sat. Oct. 10	LAB_06: VPN setup and encryption configuration (2)	☒	
Project	18 Tue. Oct. 13	Q_04 and Final Project	☒	
Project	19 Wed. Oct. 14	Final Project	☒	
Project	20 Sat. Oct. 17	Final Project	☒	
Project Due Course Procedure Manual Due	21 Tue. Oct. 20	Final Project	☒	
Review	22 Wed. Oct. 21	Final Review	☒	
Final Exam	23 Sat. Oct. 24	FINAL EXAM	☒	

E. Evaluation Plan

Evaluation type:	%	Tentative date:	Link to Ministerial elements of the competency:	✓ if part of final evaluation
Hands-on labs 6 @ 5% each)	30%		EZ03 Elements 1-4	☐
Quizzes (4 @ 2.5% each)	10%		EZ03 Elements 1-4	☐
Mid-Term Exam	10%	Class 9	EZ01 Elements 1-3	☒
Final Project	15%	Class 16	EZ01 Elements 1-4	☒
Course Procedure Manual (containing documented labs)	10%	Class 16	EZ03 Elements 1-4	☒
Final Exam	25%	Class 18	EZ03 Elements 1-4	☒
Total value:	100%			
Value of final evaluation:	60%			

F. Required Textbooks / Materials, Course Costs in addition to Texts

Title/Item:	Estimated Cost (\$):
Teacher lecture notes shared via Microsoft Teams or Moodle	N/A

G. Bibliography

Suggested books, articles, videos, websites, podcasts that can supplement the course material:
Achary, R. Cryptography and Network Security: An Introduction, Mercury Learning & Information, 2021, ISBN: 9781683926894 - e-book available at John Abbott College library online Bejtlich, R. The practice of network security monitoring: Understanding incident detection and response, No Starch Press, 1.0 edition, July 15 2013, 376 p. ISBN: 978-1593275099 Clark, B. et Alan J White. Blue team field manual (BTFM), CreateSpace Independent Publishing Platform, January 13 2017, 134 p. ISBN: 978-1541016361 Dunkerley, M. Mastering Windows security and hardening: secure and protect your Windows environment from intruders, malware attacks, and other cyber threats, Packt Publishing. 2020. ISBN: 9781839216411 - e-book available at John Abbott College library online Moyle, E. & Diana Kelley. Practical Cybersecurity Architecture : A Guide to Creating and Implementing Robust Designs for Cybersecurity Architects. Packt Publishing. 2020. ISBN 978-1838989927, 2nd Edition ISBN 978-1837637164 Nov. 10 2023 - e-book available at John Abbott College library online Tanner, N. H. Cybersecurity blue team toolkit, Wiley, April 30 2019, 288 p. ISBN-13: 978-1119552932 Tevault, D. A. Mastering Linux security and hardening: protect your Linux systems from intruders, malware attacks, and other cyber threats, Packt Publishing. 2020. ISBN: 9781838983598 - e-book available at John Abbott College library online

H. Instructional Methods:

This course will be approached from both a theoretical and practical perspective through:

- Lectures and class discussions
- In-class exercises
- Quizzes
- Individual and group work
- Hands-on work to perfect and demonstrate the desired skills
- Exams that integrate concepts and skills learned throughout the series of topics presented.

I. Program, Departmental/Discipline, and Course/Section Policies

Policy:	Description:	Indicate if Program, Department or Discipline policy
Approved departmental attendance policy	Continuing Education Policies and Guidelines (version: December 1, 2020)	Department
Late submission policy (Policy to ensure that issues relating to late submission, or resubmission, of work to be dealt with in an equitable manner)	Work submitted late may result in a 10% deduction from the grade, per calendar day	Department
Policy dealing with the expectations of classroom behaviour, including use of cell phones, laptops and other technology	Continuing Education Policies and Guidelines (version: December 1, 2020)	Department
Other course specific expectations or policies (if applicable)	Please refer to the following document concerning policies in place at the Centre for Continuing Education: Continuing Education Policies and Guidelines (version: December 1, 2020)	Department

J. College Policies

Resource:	Topic:
Policy-No.-7-IPESA-FINAL.pdf (johnabbott.qc.ca)	Student rights and responsibilities (articles 3.2 and 3.3)
	Changes to evaluation plan in the course outline (article 5.3)
	Religious holidays (articles 3.2.13 and 4.1)
	Cheating and plagiarism (articles 9.1 and 9.2)
Academic Integrity Cheating & Plagiarism Procedure (2021) • You need to log into Omnivox to access the above document • For PowerPoint on cheating and plagiarism refer to the JAC Portal: My JAC Communities / Academic Council / Curriculum Validation Committee (CVC) / Course Outlines – Reference Documents / Academic Integrity • For link to interactive tutorial on how to cite sources correctly: http://citeit.ccdmd.qc.ca	Cheating and plagiarism academic procedure and other resources
Policy 13: Policy on Student Conduct and Discipline Procedures (version: September 21, 2021)	Code of conduct